

Denis Irinyakov

MIDDLE DATA ENGINEER · DATA SCIENTIST

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Summary

Data engineer with experience in software development, data analytics and machine learning. I work at the intersection of BI and ML: designing data marts and ETL/ELT pipelines, developing data catalog practices, building engineering foundations for AI products and ensuring high data quality in the modern data stack.

I have a background in Data Science and academic research, which helps me combine AI tools, statistics and a scientific approach in engineering work. Key strengths: systematic data engineering, deep business-domain immersion, automation of routine processes, documentation and taking solutions to production operation.

I actively participate in hackathons, meetups and conferences on data engineering and data science, and I'm proactive in the work community. I develop and study new technologies to stay up to date with modern data solutions.

Work Experience

DDX Fitness

Russia

DATA ENGINEER

Oct. 2025 - Present

- Work at the intersection of **BI** and **ML** teams, developing data marts, pipelines and data engineering foundations for analytics, models and AI products.
- Orchestrate **ETL/ELT processes**: scheduled and incremental loads, data mart refreshes, product and external source integration, schedule control and SLA monitoring.
- Own **data quality**: completeness, duplicate and reconciliation checks, key data mart monitoring, alerting, incident analysis and operational fixes.
- Develop **data catalog** and metadata governance: table contracts, semantic layer, tags, review, validation and metadata sync processes.
- Contribute to the **AI trainer**: unified exercise dataset preparation, data enrichment and normalization, source matching, marts for matcher/API and ML use cases.
- Design migrations and data synchronization between analytical and product systems: idempotent loads, dry-run/apply/backfill modes, roll-back/restore and SQL result audits.
- Build internal services and APIs for expert review and operational dataset workflows: object cards, filters, coverage statistics and review verdicts.
- Build ingestion pipelines from APIs and file sources to the analytical layer: increments, SCD2/merge logic, type contracts, normalization and raw data preparation.
- Support ML workflows: feature preparation, inference processes, marts for models and analysts, reproducible calculations.
- Maintain engineering documentation, runbooks, E2E/smoke scenarios, SQL checks and MR processes to reduce operational knowledge silos.

Axenix

Saint Petersburg, Russia

DATA ENGINEER

Aug. 2024 - Oct. 2025

- Developed a modular system for dynamic generation of YAML configurations for DAGs based on Excel mappings, which reduced the time for setting up new ETL processes by 40%.
- Created a comprehensive set of development automation tools: Airflow log parser (PowerShell), synthetic data generator, artifact copying tools, and a command shell for managing Git processes, which accelerated the onboarding process for new employees and improved work quality.
- Optimized the ETL flow generator together with colleagues from other teams: implemented templating, asynchronous processing and expanded the coverage of generated artifacts.
- Automated the data quality testing process and created PL/pgSQL functions for согласующие DataFix tests, significantly reducing preparation time.
- Created and maintained stored procedures, functions, data marts and views in Greenplum for operational data quality processes.
- Provided integration of various data sources (ClickHouse, Oracle, Hive) and maintained the Data Vault 2.0 architecture.
- Organized a developer community on Telegram and GitLab to share tools and best practices.
- Developed ETL flows, prepared releases, hotfixes and data fixes.

Spacecode

Krasnodar, Russia (Remote)

DATA SCIENTIST

Apr. 2024 - Aug. 2024

- Developed a multimodal assistant based on LLM using HuggingFace models, RAG approach, LangChain and Unstructured via LLAMA_CPP/VLLM.
- Created a dynamic pricing system using interpretable regression models, which is estimated to increase segment profit by 15%.
- Used Apache Airflow to develop an ETL pipeline ensuring data freshness for ML models.
- Implemented DVC for model data versioning, which greatly simplified their tracking and result reproduction.

ASK «DirectService»

Novosibirsk, Russia (Remote)

DATA ANALYST

Sep. 2023 - Jan. 2025

- Implemented an enterprise-level Data Warehouse (DWH) from scratch on a scale of tens of terabytes of data, using PostgreSQL as the main storage and ClickHouse as the analytical storage.
- Automated the data mart update process by creating a comprehensive pipeline based on Shell scripts and Windows Task Scheduler.
- Set up a data migration mechanism from ClickHouse to PostgreSQL, ensuring uninterrupted operation of analytical systems.
- Together with system analysts, developed methodologies for monitoring and optimizing DWH performance.
- Set up monitoring based on Postgres extensions, CI/CD for verified code delivery using Flyway and Github Actions.

LLC «ColorDent»

Moscow, Russia (Remote)

ML ENGINEER

Jun. 2023 - Dec. 2023

- Developed an ML image classifier for tooth disease diagnosis using OpenCV and TensorFlow Lite. The model achieved accuracy of over 95% on test data.
- Integrated ML models into a mobile application using MLflow for model version management and DVC for data versioning.
- Optimized model performance for operation on resource-constrained devices, which allowed it to be used on mobile platforms without significant quality loss.
- Developed a system for tracking model training metrics and implemented automated testing to ensure high solution quality.

Main Education

FRC Biodiversity of FEB RAS

Vladivostok, Russia

GRADUATE SCHOOL

2019 - 2023

- Specialization - Ecology. Deep data analysis, statistics, mathematical modeling of ecological processes.
- Working with spatial and time series for ecological data analysis.
- Using the R programming language, statistical methods and mathematical modeling.
- Publication of scientific articles in the field of ecology and data analysis.

Data education

School of Programming 21

Novosibirsk, Russia

FOUNDATION - SCHOOL 21

2023 - 2025

- C, bash, Python, SQL, Go, Ruby, algorithms, data structures.
- Deep dive into programming languages and development methodologies.
- Peer-to-peer learning and project activities.

Yandex Practicum

Online

DATA ANALYST

Feb. 2023 - 2024

- Python libraries: pandas, scipy, numpy, matplotlib, scikit-learn, plotly, sqlalchemy.
- SQL queries, data analysis, machine learning.
- Statistical analysis and data visualization.

T1 Data Engineer School

Online

DATA ENGINEER

2024 - Jun. 2024

- Data engineer school from T1 holding with a graduation project.
- ETL/ELT processes, data architecture, modern tools.
- Practical projects for processing big data in Greenplum.

Arenadata

Online

ARENADATA HADOOP ADMINISTRATOR

Jan. 2025 - Mar. 2025

- Operation of Arenadata Hadoop. HDFS, Zookeeper, Grafana, Spark, Kafka.
- Monitoring services of the Hadoop ecosystem.
- Deploying clusters in the Hadoop ecosystem.

Specialized Training

School 21

C/C++ DEVELOPMENT

- Writing standard libraries in C, data structures, algorithms.
- Memory management, writing desktop programs in C++ & QT.
- Projects: 3dViewer, Calculator, Tetris and others.

Novosibirsk, Russia

2023 - 2024

School 21

PYTHON DEVELOPMENT

- Dockerfile, Makefile, Ansible, Neo4j, Python.
- unittest, pytest, subprocess, bs4, dotenv, scipy, plotly.
- grpc, sqlalchemy, asyncio, fastapi, aiogram.

Novosibirsk, Russia

2024

School 21

SQL & DATABASE

- pl/pgsql, functions, procedures, window functions.
- Subqueries, CTE, nested queries, recursive queries.
- Database performance optimization.

Novosibirsk, Russia

2024.1 - 2024.2

School 21

DEVOPS & INFRASTRUCTURE

- Linux, VM, Docker, CI/CD, bash.
- Prometheus, Grafana, Makefile, Docker, Vagrantfile.
- Containerization and orchestration of applications, CI/CD setup.

Novosibirsk, Russia

2024.3 - 2024.4

Key Projects & Achievements

School 21

DATA SCIENCE CLUB LEADER

- Formed a community within School 21 of 60+ people.
- Conducted educational sessions in a peer-to-peer style.
- Joint participation in hackathons, developing team soft skills.
- Team leadership at hackathons and coordination of project activities.

Novosibirsk, Russia

Mar. 2023 - Sep. 2024

SmartSocialHack

HACKATHON WINNER

- First place in a case dedicated to socially significant problems.
- Using LLM, PostgreSQL, FastAPI to create a solution.
- Developing a mobile and web application as a team.
- Integration of PWA technologies to create a comprehensive solution.
- Development based on microservice architecture.

Moscow, Russia

June. 2024 - Jul. 2024

Technical Skills

System Fundamentals	Linux, WSL, grep, env, Makefile, cron, systemd, git, ssh, rsync
Programming Languages	Python, SQL + PL/pgSQL, C, Go, Shell/Bash
Favorite Tools	ClickHouse, OpenMetadata, dbt, Airflow, PostgreSQL, Docker, GitLab
Data Storage & Databases	PostgreSQL, ClickHouse, Greenplum, MySQL, SQLite, Neo4j, ChromaDB, S3, Iceberg, Delta Lake, OpenMetadata
Python Stack	uv, pip, pandas, polars, scipy, numpy, matplotlib, scikit-learn, CatBoost, plotly, SQLAlchemy, asyncio, FastAPI, Flask, Pytest
DevOps & Infrastructure	Docker, Docker Compose, CI/CD, YAML, GitHub Actions, GitLab Runner, Vagrant, Ansible, Consul, Prometheus, Grafana
Data Orchestration	Apache Airflow, dlt, PeerDB, Kafka, Redis, Spark, Trino, Flink, dbt, FDW, pg_cron
Data Architecture	Data Lake, Lakehouse, DWH, Data Catalog, Metadata Management, Data Contracts, Semantic Layer, Data Vault 2.0, Ancho, Kimball, scd1/2/3
Machine Learning	ML and data analysis, TensorFlow Lite, CV, NLP, LLM, Time Series, EDA, Feature Engineering, Data Cleaning, Data Augmentation, Data Visualization, statistics
AI Workflows & Integration	RAG, Vector DB, MCP, n8n, Copilot, Cursor, OpenRouter, HuggingFace, OpenAI, Ollama

Soft Skills & Business Skills

Leadership & Teamwork	Teamwork, Leadership, Kanban, Agile
Problem Solving	Critical thinking, data analysis, hypothesis generation, A/B testing
Adaptability	Adaptability, high learning ability, following technology trends
Innovation	Innovative thinking, using AI tools, DevOps, modern DB architecture approaches
Data Quality	Data quality, integrity, accuracy, data availability at all stages of the life cycle
Security & Standards	Information security, development standards